


```
In [5]: import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
from scipy import stats

csv_url = 'https://archive.ics.uci.edu/ml/machine-learning-database

col_names = ['Sepal_Length', 'Sepal_Width', 'Petal_Length', 'Petal_Wid

iris = pd.read_csv(csv_url, names=col_names)

print(iris.head(30))
```

	Sepal_Length	Sepal_Width	Petal_Length	Petal_Width	Spec
ies					
0	5.1	3.5	1.4	0.2	Iris-set
osa					
1	4.9	3.0	1.4	0.2	Iris-set
osa					
2	4.7	3.2	1.3	0.2	Iris-set
osa					
3	4.6	3.1	1.5	0.2	Iris-set
osa					
4	5.0	3.6	1.4	0.2	Iris-set
osa					
5	5.4	3.9	1.7	0.4	Iris-set
osa					
6	4.6	3.4	1.4	0.3	Iris-set
osa					
7	5.0	3.4	1.5	0.2	Iris-set
osa					
8	4.4	2.9	1.4	0.2	Iris-set
osa					
9	4.9	3.1	1.5	0.1	Iris-set
osa					
10	5.4	3.7	1.5	0.2	Iris-set
osa					
11	4.8	3.4	1.6	0.2	Iris-set
osa					
12	4.8	3.0	1.4	0.1	Iris-set
osa					
13	4.3	3.0	1.1	0.1	Iris-set
osa					
14	5.8	4.0	1.2	0.2	Iris-set
osa					
15	5.7	4.4	1.5	0.4	Iris-set
osa					
16	5.4	3.9	1.3	0.4	Iris-set
osa					
17	5.1	3.5	1.4	0.3	Iris-set
osa					
18	5.7	3.8	1.7	0.3	Iris-set
osa					
19	5.1	3.8	1.5	0.3	Iris-set
osa					
20	5.4	3.4	1.7	0.2	Iris-set
osa					
21	5.1	3.7	1.5	0.4	Iris-set
osa					
22	4.6	3.6	1.0	0.2	Iris-set
osa					
23	5.1	3.3	1.7	0.5	Iris-set
osa					
24	4.8	3.4	1.9	0.2	Iris-set
osa					
25	5.0	3.0	1.6	0.2	Iris-set
osa					
26	5.0	3.4	1.6	0.4	Iris-set
osa					
27	5.2	3.5	1.5	0.2	Iris-set
osa					
28	5.2	3.4	1.4	0.2	Iris-set
osa					

29	4.7	3.2	1.6	0.2	Iris-set
osa					

```
In [6]: mean_values = iris[['Sepal_Length', 'Sepal_Width', 'Petal_Length',  
print("\nMean values:")  
print(mean_values)
```

```
Mean values:  
Sepal_Length    5.843333  
Sepal_Width     3.054000  
Petal_Length    3.758667  
Petal_Width     1.198667  
dtype: float64
```

```
In [7]: median_values = iris[['Sepal_Length', 'Sepal_Width', 'Petal_Length']  
print("\nMedian values:")  
print(median_values)
```

```
Median values:  
Sepal_Length    5.843333  
Sepal_Width     3.054000  
Petal_Length    3.758667  
Petal_Width     1.198667  
dtype: float64
```

```
In [8]: mode_values = iris[['Sepal_Length', 'Sepal_Width', 'Petal_Length',  
print("\nMode values:")  
print(mode_values)
```

```
Mode values:  
Sepal_Length    5.0  
Sepal_Width     3.0  
Petal_Length    1.5  
Petal_Width     0.2  
Name: 0, dtype: float64
```

```
In [13]: min_val = iris[['Sepal_Length', 'Sepal_Width', 'Petal_Length', 'Petal_Width']]
max_val = iris[['Sepal_Length', 'Sepal_Width', 'Petal_Length', 'Petal_Width']]
range_values = max_val - min_val

print("\nRange values (Max - Min):")
print(range_values)
```

```
Range values (Max - Min):
Sepal_Length    3.6
Sepal_Width     2.4
Petal_Length    5.9
Petal_Width     2.4
dtype: float64
```

```
In [11]: variance_values = iris[['Sepal_Length', 'Sepal_Width', 'Petal_Length', 'Petal_Width']]
print("\nVariance values:")
print(variance_values)
```

```
Variance values:
Sepal_Length    0.685694
Sepal_Width     0.188004
Petal_Length    3.113179
Petal_Width     0.582414
dtype: float64
```

```
In [12]: std_deviation_values = iris[['Sepal_Length', 'Sepal_Width', 'Petal_Length', 'Petal_Width']]
print("\nStandard Deviation values:")
print(std_deviation_values)
```

```
Standard Deviation values:
Sepal_Length    0.828066
Sepal_Width     0.433594
Petal_Length    1.764420
Petal_Width     0.763161
dtype: float64
```

```
In [14]: summary_stats = iris[['Sepal_Length', 'Sepal_Width', 'Petal_Length']

print("\nSummary Statistics (Describe):")
print(summary_stats)
```

```
Summary Statistics (Describe):
```

	Sepal_Length	Sepal_Width	Petal_Length	Petal_Width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.054000	3.758667	1.198667
std	0.828066	0.433594	1.764420	0.763161
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

```
In [ ]:
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